The eigenproblem

For $A \in \mathbb{C}^{n \times n}$, suppose a scalar λ and nonzero vector x satisfy

$$Ax = \lambda x,$$

Then λ is an eigenvalue and x an eigenvector.

Equivalently,

$$(A - \lambda I)x = 0.$$

Since $x \neq 0$, the matrix $A - \lambda I$ is singular. Hence

$$p_A(\lambda) = \det(\lambda I - A) = 0.$$

The one-dimensional subspace $\operatorname{span}\{x\}$ is invariant. Over the reals, a negative eigenvalue reverses the arrow direction while preserving the line.

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Characteristic polynomial, trace, and determinant

Using the monic convention for an $n \times n$ matrix,

$$p_A(\lambda) = \lambda^n - a_1\lambda^{n-1} + \cdots + (-1)^n \det A.$$

Counting eigenvalues with algebraic multiplicity gives

$$\sum_i \lambda_i = \operatorname{tr} A, \qquad \prod_i \lambda_i = \det A.$$

For example,

$$A = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}.$$

The trace is 6 and determinant is 8.

$$p_A(\lambda) = \lambda^2 - 6\lambda + 8 = (\lambda - 4)(\lambda - 2).$$

The eigenvalues are 4 and 2. Trace and determinant check the spectrum but do not generally determine it in dimensions above two.

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How operations transform eigenvalues

If $Ax = \lambda x$, then for nonnegative integers k and polynomials q,

$$A^k x = \lambda^k x,$$

$$A^{-1} x = \lambda^{-1} x \quad (A \text{ invertible}),$$

$$q(A)x = q(\lambda)x.$$

Shift:

$$(A + cI)x = (\lambda + c)x.$$

Eigenspace:

$$E_\lambda = N(A - \lambda I).$$

The eigenspace is a subspace whose dimension is the geometric multiplicity. A triangular matrix has its diagonal entries as eigenvalues, counted with algebraic multiplicity.

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Algebraic and geometric multiplicities

The multiplicity of a root in the characteristic polynomial is its algebraic multiplicity a_λ. The quantity

$$g_\lambda = \dim N(A - \lambda I)$$

is its geometric multiplicity. Extend an eigenspace basis to a basis of the whole space. The resulting matrix is block upper triangular with block λI_{g_λ}, so its characteristic polynomial contains $(t - \lambda)^{g_\lambda}$. Therefore

$$1 \leq g_\lambda \leq a_\lambda.$$

If any eigenvalue has $g_\lambda < a_\lambda$, the matrix is defective.

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Algebraic and geometric multiplicities: an example

For example,

$$A = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}.$$

The algebraic multiplicity is 2. Solving

$$\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} x = 0$$

gives only $x_2 = 0$, hence

$$E_2 = \text{span}\{(1, 0)^T\}, \qquad g_2 = 1.$$

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A complete two-dimensional example

$$A = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}.$$

$\lambda = 4$:

$$\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} x = 0,$$

Choose $x_1 = (1, 1)^T$.

$\lambda = 2$:

$$\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} x = 0,$$

Choose $x_2 = (1, -1)^T$.

These eigenvectors are orthogonal because A is symmetric. If

$$u_0 = c_1 x_1 + c_2 x_2,$$

Then

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A complete two-dimensional example (continued)

$$u_k = A^k u_0 = c_1 4^k x_1 + c_2 2^k x_2.$$

For $c_1 \neq 0$, the eigenvalue-4 mode determines the limiting direction.

Taking $c_1 = c_2 = 1$,

$$u_0 = x_1 + x_2 = (2, 0)^T, \quad u_1 = (6, 2)^T, \quad u_2 = (20, 12)^T.$$

The diagram shows $c_1 4^k x_1$, $c_2 2^k x_2$, and their sum u_k. For $c_1 = 0, c_2 = 1$, instead,

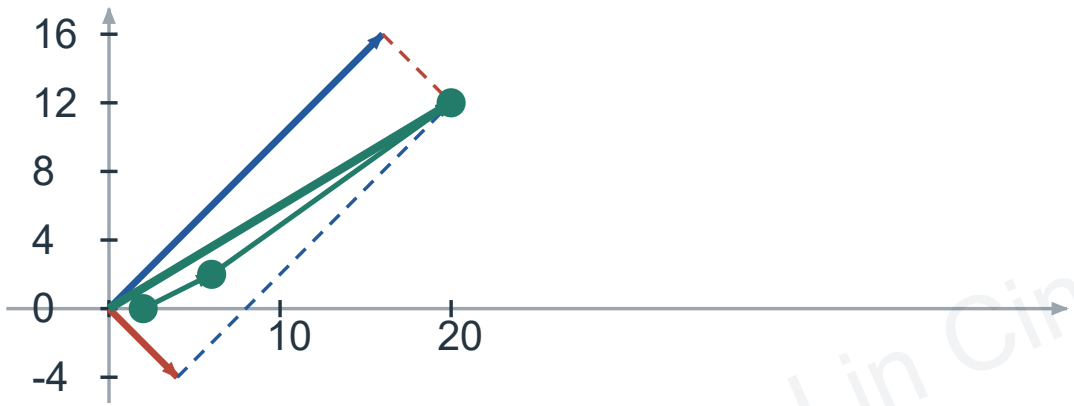
$$u_0 = (1, -1)^T, \quad u_1 = (2, -2)^T, \quad u_2 = (4, -4)^T,$$

the trajectory stays in the eigenvector direction x_2. It has no component in the larger-eigenvalue mode. Real coefficients can also be used with unit eigenvectors; here the diagram retains the specified nonunit x_1, x_2, so its coefficients refer to that basis.

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Numerical evolution of eigenmodes

Eigenmodes: components, sum, and trajectory



A=[[3,1],[1,3]]

$\lambda_1=4$ · $x_1=(1,1)$

$\lambda_2=2$ · $x_2=(1,-1)$

Blue: $4^2x_1=(16,16)$

Red: $2^2x_2=(4,-4)$

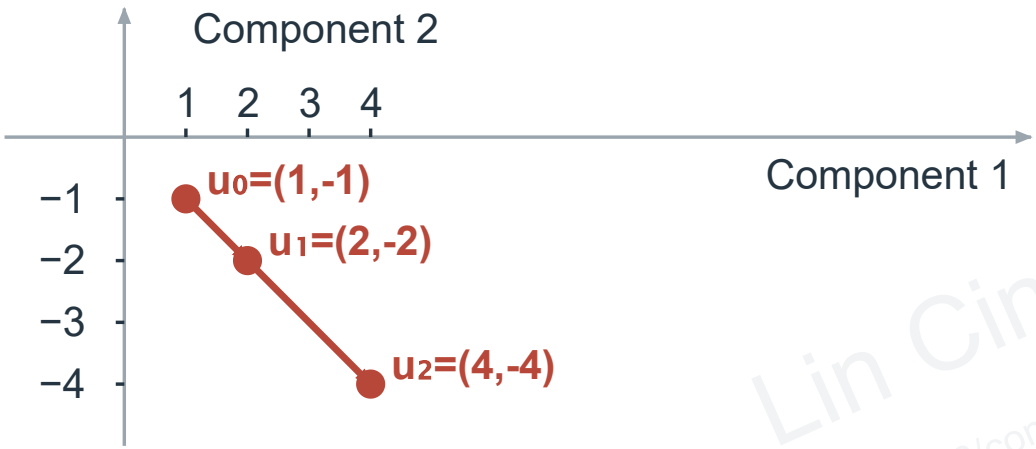
Green: $u_2=(20,12)$

$u_0,u_1,u_2=(2,0),(6,2),(20,12)$

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Dominant direction and the exceptional initial state

Exceptional trajectory $c_1=0$: always along $x_2=(1,-1)$



$c_1=0$, $c_2=1$

$u_k=2^k(1,-1)$

Every u_k has the same eigenvector direction.

Fixed ratio; no 4^k dominant term.

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Rotation and complex eigenvalues

$$R_\theta = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}.$$

$$p(\lambda) = \lambda^2 - 2\cos\theta\,\lambda + 1,$$

$$\lambda = e^{\pm i\theta}.$$

Unless $\theta \equiv 0 \pmod{\pi}$, a planar rotation has no nonzero real eigenvector.

Complex eigenvalues of a real matrix come in conjugate pairs. If

$$Ax = \lambda x,$$

taking complex conjugates gives

$$A\overline{x} = \overline{\lambda}\,\overline{x}.$$

Real solutions require compatible conjugate coefficients. A single complex mode is generally not a real physical solution.

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Diagonalization

Suppose A has n independent eigenvectors x_i. Form

$$X = [x_1 \ \cdots \ x_n].$$

$$AX = X\Lambda, \qquad \Lambda = \operatorname{diag}(\lambda_i),$$

$$A = X\Lambda X^{-1}, \qquad X^{-1}AX = \Lambda.$$

Diagonalization is equivalent to the eigenspace dimensions summing to n. Eigenvectors for distinct eigenvalues are independent. Indeed, a shortest nontrivial relation $\sum c_i x_i = 0$, after applying $A - \lambda_k I$, would give a shorter relation, a contradiction. Thus n distinct eigenvalues suffice. Repeated eigenvalues can also have complete eigenspaces, as for $2I$.

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Matrix powers and discrete-time stability

For $A = X\Lambda X^{-1}$, we have $A^k = X\Lambda^k X^{-1}$. If $u_0 = \sum_j c_j x_j$, the recurrence $u_{k+1} = Au_k$ has solution $u_k = \sum_j c_j \lambda_j^k x_j$. This eigenvector expansion requires diagonalizability.

The s×s Jordan block $J_s(\lambda) = \lambda I_s + N_s$ has λ on its diagonal, 1 directly above it, and 0 elsewhere; thus $N_s^s = 0$. Use these blocks for a general square matrix. If $J = \lambda I + N$, $N^m = 0$, and $\lambda \neq 0$, the binomial theorem gives

$$J^k = \sum_{j=0}^{\min(k,m-1)} \binom{k}{j} \lambda^{k-j} N^j.$$

When $0 < |\lambda| < 1$, each coefficient is a polynomial of degree at most j times geometric decay, hence tends to zero. When $\lambda = 0$, simply use $J^k = 0$ for $k \geq m$, without negative powers.

Thus $A^k \to 0$ exactly when $\rho(A) = \max_j |\lambda_j| < 1$. Necessity follows on an eigenvector: if $|\lambda| \geq 1$, then $A^k x = \lambda^k x$ does not tend to zero.

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What else matters on the unit circle?

Eigenvalues and Jordan structure	Long-term behavior of A^k
All eigenvalue moduli below 1	Tends to zero, even if defective
All moduli at most 1; unit-circle blocks have size 1	Bounded; any unit-circle mode prevents decay to zero
A unit-circle block has size greater than 1	Polynomial growth on some generalized eigenvectors
An eigenvalue has modulus greater than 1	Exponential growth on its eigenvector

For example I preserves the initial state, but $J = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ gives $J^k e_2 = (k, 1)^T$. Both have only eigenvalue 1; block size distinguishes them. Rotation powers are bounded, but an angle that is an irrational multiple of 2π has no common finite period.

These are long-term statements. Even $\rho(A) < 1$ permits finite-time growth; eigenvalues alone do not imply a decreasing Euclidean norm.

Similarity invariants

If

$$C = B^{-1}AB,$$

then

$$C - \lambda I = B^{-1}(A - \lambda I)B,$$

Thus characteristic polynomial, eigenvalues, trace, determinant, rank, and Jordan structure are unchanged.

If $Ax = \lambda x$, set $y = B^{-1}x$. Then

$$Cy = \lambda y.$$

Similarity changes coordinates while preserving the intrinsic spectrum.

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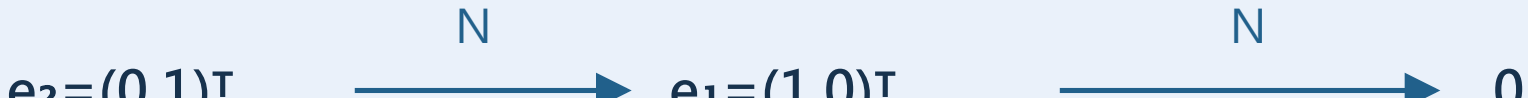
Jordan chains supply the missing direction

Take $A = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix} = 2I + N$. The equation $(A - 2I)x = 0$ gives only $x_2 = 0$: its eigenspace has dimension 1, so A is not diagonalizable. Put $e_1 = (1, 0)^T, e_2 = (0, 1)^T$.

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Mappings in a Jordan chain

The chain arrows apply N, not A



$N^2e_2=0$ but $Ne_2\neq0$, so the chain has length 2.

Apply A to e_1

$Ae_1 = 2e_1$

Eigenvector

Apply A to e_2

$Ae_2 = e_1 + 2e_2$

Generalized eigenvector; image is not parallel

$A=2I+N$, $N=[[0,1],[0,0]]$. The basis (e_1,e_2) contains only one eigenvector.

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Where does the polynomial factor in a Jordan power come from?

Since $N^2 = 0$ and $2I$ commutes with N, only two terms survive the binomial expansion. For integers $k \geq 0$,

$$A^k = 2^k I + k 2^{k-1} N = \begin{bmatrix} 2^k & k2^{k-1} \\ 0 & 2^k \end{bmatrix}.$$

Take $u_0 = e_2$ and write $u_k = (a_k, b_k)^T$. Matrix multiplication gives $a_{k+1} = 2a_k + b_k$, $b_{k+1} = 2b_k$, so $b_k = 2^k$ and

$$a_k = \sum_{j=0}^{k-1} 2^{k-1-j} b_j = \sum_{j=0}^{k-1} 2^{k-1} = k2^{k-1} \quad (k \geq 1).$$

At k=0 the empty sum gives a_0=0. Every step injects another e_1 component; k such contributions produce the polynomial factor.

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Matrix multiplication and the closed form give the same trajectory

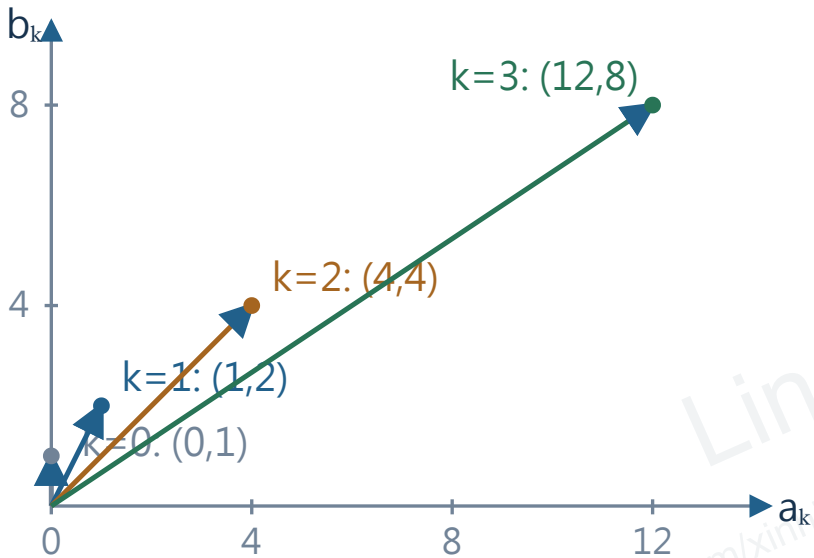
Use the same A with $u_0 = e_2$. Each arrow starts at the origin and ends at $A^k e_2$.

Check $A(4, 4)^T = (12, 8)^T$. Replacing A by 2I would give $(0, 2^k)^T$, losing the entire horizontal component.

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Stepwise evolution of a Jordan block

Track $u_k = A^k e_2$ on axes with equal unit scales



k	a_k	b_k
0	0	1
1	1	2
2	4	4
3	12	8

$a_{k+1} = 2a_k + b_k$; $b_{k+1} = 2b_k$

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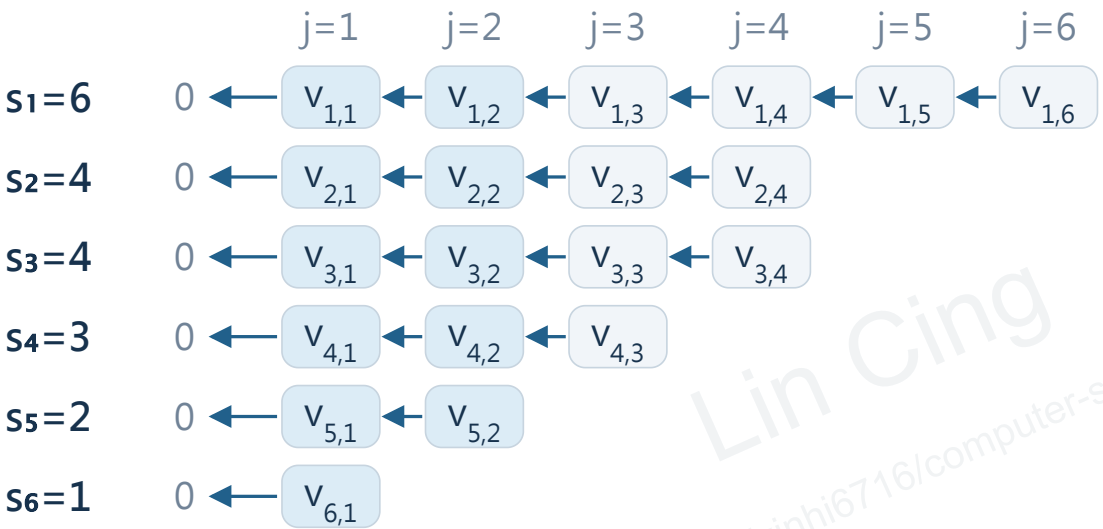
Recovering Jordan blocks from kernel dimensions

Let N be nilpotent, with chain lengths $s_1, \dots, s_r$. Set $d_k = \dim \ker N^k$, $d_0 = 0$. Each chain contributes precisely its first $\min(k, s_i)$ vectors to the kernel of N^k.

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Nullities of a nilpotent matrix

N sends each cell one position left; the first goes to 0



$k=2$

$\dim \ker(N^2)$

$=2+2+2+2+2+1$

$=11$

First $\min(k,s)$ cells per chain lie in $\ker(N^k)$.

Blue cells form a basis of $\ker(N^2)$; each row is a separate Jordan block.

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What does each nullity increment count?

A chain of length at least k contributes one new basis vector between $\ker N^{k-1}$ and $\ker N^k$. Therefore

$$d_k - d_{k-1} = \#\{i : s_i \geq k\}, \qquad \#\{i : s_i = k\} = 2d_k - d_{k-1} - d_{k+1}.$$

For $(d_0, \ldots, d_6) = (0, 6, 11, 15, 18, 19, 20)$, with d_7=20, the increments are (6,5,4,3,1,1,0). Adjacent differences give block counts (1,1,1,2,0,1) for lengths 1 through 6. Thus the lengths are 6,4,4,3,2,1, totaling 20 dimensions.

Checks. The value d_1=6 counts blocks. The first full nullity d_6=20 identifies maximum block length 6, so N^6=0 but $N^5 \neq 0$. For a general eigenvalue λ, apply the same method to N=A−λI on its generalized eigenspace.

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From nullity to Jordan chains: a five-dimensional example

Consider the matrix below with the convention $p_A(t) = \det(tI - A)$:

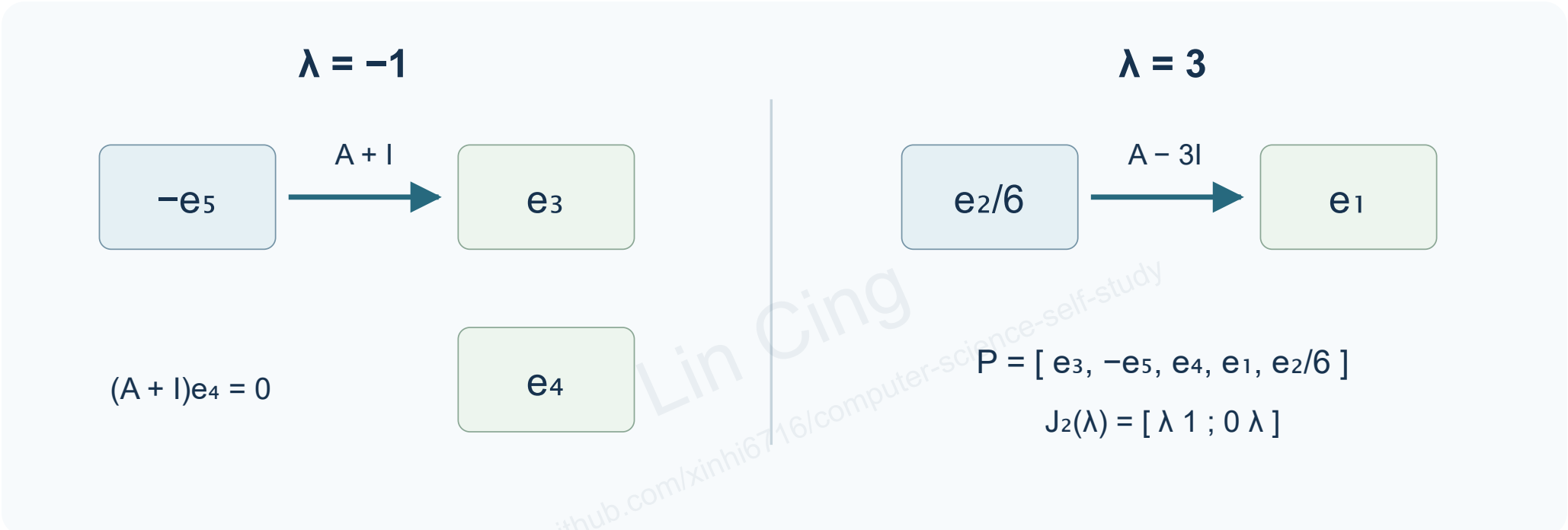
$$A = \begin{bmatrix} 3 & 6 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & -1 \\ 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & -1 \end{bmatrix}, \quad p_A(t) = (t-3)^2(t+1)^3.$$

Using $\det(A - tI)$ instead changes the overall sign in dimension five, but not the roots or block sizes.

For $\lambda = -1$, the nullities of $(A + I)^k$ at $k = 1, 2, 3$ are $2, 3, 3$, giving blocks of sizes 2 and 1. For $\lambda = 3$, the nullities of $(A - 3I)^k$ are $1, 2, 2$, giving one block of size 2.

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Find the chain vectors and check their images



$(A+I)(-e_5) = e_3$ and $(A+I)e_3 = (A+I)e_4 = 0$. The other chain satisfies $(A-3I)(e_2/6) = e_1$, $(A-3I)e_1 = 0$. Placing each eigenvector before its generalized vector produces a Jordan block with 1 on its superdiagonal.

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Chain order determines the Jordan form

Let $P = [e_3, -e_5, e_4, e_1, e_2/6]$. Its columns are independent and $\det P = 1/6 \neq 0$. Applying the chain relations column by column gives

$$P^{-1}AP = \operatorname{diag}\left(\begin{bmatrix} -1 & 1 \\ 0 & -1 \end{bmatrix}, [-1], \begin{bmatrix} 3 & 1 \\ 0 & 3 \end{bmatrix}\right).$$

For example, $A(-e_5) = -(-e_5) + e_3$, so its coordinate column is $(1, -1, 0, 0, 0)^T$. Reversing every chain moves the 1s below the diagonal; the forms are similar, but the column order of P must change accordingly.

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Spectral decomposition

If the rows of X^{-1} are y_j^T,

$$A = \sum_{j=1}^{n} \lambda_j x_j y_j^T .$$

Each $x_j y_j^T$ has rank one. For a real symmetric matrix choose a real orthogonal matrix Q:

$$A = Q \Lambda Q^T = \sum_j \lambda_j q_j q_j^T .$$

This scales independent orthonormal directions. Nonnormal matrices generally have nonorthogonal eigenvectors, so eigenvalues alone do not describe transient amplification.

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From scalar to matrix ODEs

Scalar:

$$u' = \lambda u \quad \Longrightarrow \quad u(t) = Ce^{\lambda t}.$$

System:

$$u' = Au, \qquad u(0) = u_0.$$

If A is diagonalizable,

$$u(t) = \sum_j c_j e^{\lambda_j t} x_j.$$

$$c = X^{-1} u_0.$$

For $\lambda = \alpha + i\beta$,

$$e^{\lambda t} = e^{\alpha t}(\cos \beta t + i \sin \beta t).$$

The real part determines exponential growth or decay. The imaginary part determines oscillation.

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The matrix exponential

$$e^{At} = I + At + \frac{A^2 t^2}{2!} + \cdots .$$

The absolutely convergent power series can be differentiated term by term on bounded time intervals.

$$\frac{d}{dt} e^{At} = A e^{At} = e^{At} A.$$

Diagonalizable:

$$e^{At} = X e^{\Lambda t} X^{-1} = \sum_j e^{\lambda_j t} x_j y_j^T.$$

Defective $A = 2I + N$, $N^2 = 0$:

$$e^{At} = e^{2t}(I + tN).$$

Eigenvalue real parts determine asymptotic stability. Nonnormal transient behavior also depends on norms or more detailed information about the matrix.

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Stability and its boundary cases

$$\begin{bmatrix} u_1' \\ u_2' \end{bmatrix} = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}.$$

Use eigenbasis：

$$u(t) = c_1 e^{4t}(1,1)^T + c_2 e^{2t}(1,-1)^T.$$

Both eigenvalues are positive, so every nonzero initial vector produces a growing solution.

The zero solution of a linear system is asymptotically stable exactly when every eigenvalue has negative real part. A Jordan block $J = \lambda I + N$ of size m gives $e^{Jt} = e^{\lambda t} \sum_{\ell=0}^{m-1} t^\ell N^\ell / \ell!$. A negative real part makes every polynomial times the decaying exponential tend to zero. A positive real part gives an unstable mode.

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Stability and its boundary cases (continued)

At the boundary, if all real parts are nonpositive and every imaginary-axis Jordan block has size one, e^{At} is bounded and the zero solution is Lyapunov stable. Any remaining imaginary-axis mode prevents asymptotic stability. For $A = 0$, solutions retain their initial values; for skew-symmetric rotation, they retain their length. In contrast, $J = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ gives $e^{Jt} = I + tJ$, taking e_2 to $(t, 1)^T$. Zero real parts can therefore still be unstable. These conclusions concern the stated linear system.

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Asymptotic stability still permits initial norm growth

Let $B = \begin{bmatrix} -1 & 4 \\ 0 & -1 \end{bmatrix}$, $u_0 = (1,1)^T/\sqrt{2}$. Both eigenvalues are −1, but

$$u(t) = e^{-t}(1+4t,1)^T/\sqrt{2}, \qquad g(t) = \|u(t)\|_2 = e^{-t}\sqrt{\frac{(1+4t)^2+1}{2}}.$$

Put $h(t) = g(t)^2$. Differentiation gives

$$h'(t) = e^{-2t}\{4(1+4t) - (1+4t)^2 - 1\}, \qquad h'(0) = 2 > 0.$$

Thus this unit initial state grows first. With z=1+4t, stationarity gives z^2−4z+1=0. Only z=2+$\sqrt{}$3 corresponds to t≥0, so $t_* = (1+\sqrt{3})/4 \approx 0.683$, $g(t_*) \approx 1.379$. All components eventually decay because $te^{-t} \to 0$.

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Why can the same eigenvalues give different curves?

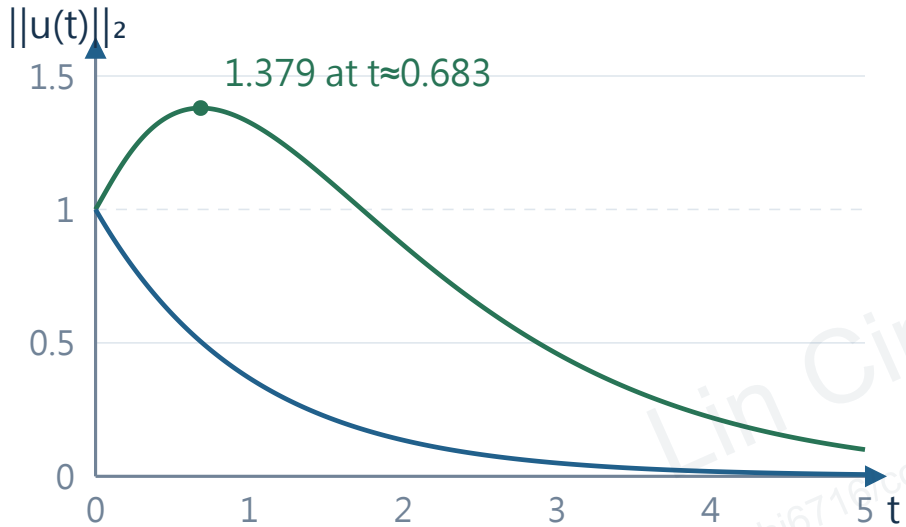
Compare B with C=−I, using the same unit initial state $u_0 = (1,1)^T/\sqrt{2}$. For C the norm is exactly e^{-t}. For B the off-diagonal term feeds the second component into the first.

The peak is for the **specified initial state**, $\|e^{Bt}u_0\|_2$, not the maximum over unit states $\|e^{Bt}\|_2$. With initial state e_1, B also gives pure e^{-t} decay.

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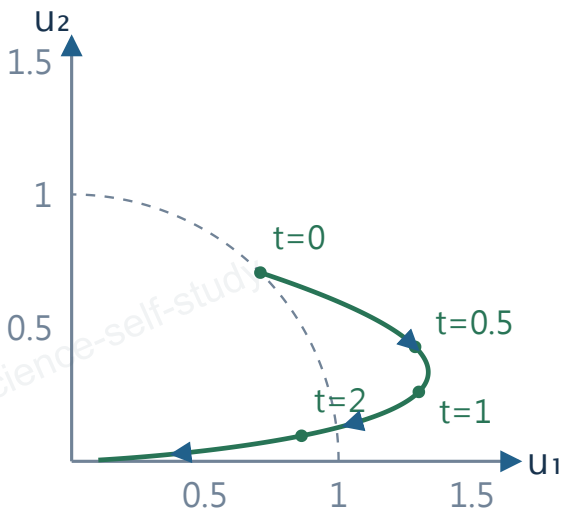
Transient growth and long-term decay

Same eigenvalues −1,−1 and unit initial state; different transient norms



B=[[-1,4],[0,-1]]

C=−I



B trajectory; dashed unit circle

Equal unit scales on both axes.

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The symmetric part tests contraction for every initial state

For a real system u′=Bu,

$$\frac{d}{dt}\|u\|_2^2 = u^T(B+B^T)u = 2u^THu, \qquad H=(B+B^T)/2.$$

Every initial state has a nonincreasing norm exactly when H is negative semidefinite. Sufficiency follows along each solution. For necessity, evaluate the derivative at t=0 for arbitrary initial vectors. The complex version uses conjugate transpose.

Here $H=\begin{bmatrix}-1 & 2\\ 2 & -1\end{bmatrix}$ has eigenvalues 1,−3. Our u_0 is a unit eigenvector for 1, giving initial energy derivative 2. In contrast, the three-dimensional matrix in Exercise 8 has maximum H-eigenvalue $-1+\sqrt{2}/2<0$, so every nonzero solution strictly contracts. **Neither defectiveness nor nonnormality alone proves transient amplification.**

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Converting a second-order ODE

$$u'' + Bu' + Cu = 0.$$

For scalar constants B, C, set $v = u'$.

$$\frac{d}{dt}\begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -C & -B \end{bmatrix}\begin{bmatrix} u \\ v \end{bmatrix}.$$

Characteristic polynomial:

$$\det\begin{bmatrix} -\lambda & 1 \\ -C & -B-\lambda \end{bmatrix} = \lambda^2 + B\lambda + C.$$

Complex conjugate eigenvalues produce oscillation and their real parts determine decay. The initial data $(u(0), u'(0))$ become a two-dimensional initial vector.

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Real eigenvalues of real symmetric matrices

Let $S \in \mathbb{R}^{n \times n}$, $S = S^T$, and choose a nonzero $x \in \mathbb{C}^n$ with $Sx = \lambda x$. Real entries and symmetry give $S^* = \overline{S}^T = S$. Using the conjugate transpose,

$$x^* S x = \lambda x^* x.$$

because

$$\overline{x^* S x} = x^* S^* x = x^* S x,$$

the numerator is real. Since $x^* x > 0$,

$$\lambda = \frac{x^* S x}{x^* x} \in \mathbb{R}.$$

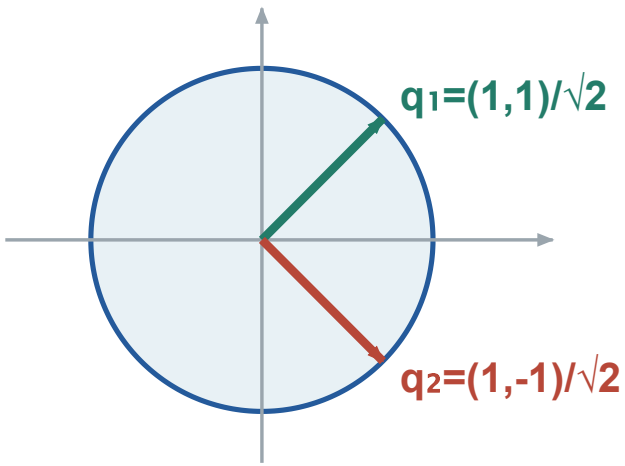
The first equality uses conjugation of a scalar; the second uses $S^* = S$, which does not follow from $S^T = S$ for a complex matrix. The denominator $x^* x = \sum_j |x_j|^2 > 0$ establishes the conclusion.

The counterexample $S = [i]$ satisfies $S^T = S$ but $S^* = [-i] \neq S$, with eigenvalue i. The complex-space premise is Hermitian symmetry $S^* = S$. The same proof gives real eigenvalues, and diagonalization uses a unitary matrix $S = U \Lambda U^*$.

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Real symmetric matrices

Orthonormal eigenbasis: $Q^TSQ=\text{diag}(4,2)$



$S=[[3,1],[1,3]]$

$Q=(1/\sqrt{2})[[1,1],[1,-1]]$

$Q^TSQ=\text{diag}(4,2)$

$S=Q\Lambda Q^T$; the axes are perpendicular.

Normalize (1,1),(1,-1) before forming Q.

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Complex symmetric versus Hermitian

Complex symmetric does not imply Hermitian

$S=[i]$ · $S^T=S$

Only eigenvalue: i

$S^*=[-i]\neq S$

The Hermitian premise is necessary.

Hermitian : $S^*=S$

$x^*Sx=\lambda x^*x$

$x^*x>0 \Rightarrow \lambda\in R$

$S=U\Lambda U^*$, with U unitary

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Orthogonality of distinct eigenspaces

Let $S \in \mathbb{R}^{n \times n}$ be real symmetric, with real eigenvectors x, y. If $Sx = \lambda x$, $Sy = \mu y$,

$$\lambda x^T y = (Sx)^T y = x^T S^T y = \mu x^T y.$$

Consequently,

$$(\lambda - \mu) x^T y = 0.$$

then $\lambda \neq \mu$ implies $x \perp y$. A complete orthonormal eigenbasis follows by induction. The continuous quadratic form $x^T S x$ attains a maximum on the unit sphere; its extremum condition gives a real unit eigenvector q. For $v \perp q$, $q^T S v = (Sq)^T v = 0$, so $q^\perp$ is invariant. The restriction remains symmetric. Induction gives a complete orthonormal eigenbasis; within repeated eigenspaces, use Gram–Schmidt.

Spectral theorem：

$$S = Q \Lambda Q^T.$$

For example,

$$S = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}, \qquad Q = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}.$$

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Orthogonality of distinct eigenspaces (continued)

Since $Q^T Q = I$ and

$$Q^T S Q = \begin{bmatrix} 4 & 0 \\ 0 & 2 \end{bmatrix},$$

we obtain $S = Q \Lambda Q^T$. The unnormalized vectors $(1, 1)^T, (1, -1)^T$ specify directions only. Divide them by $\sqrt{2}$ before using them as columns of Q.

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Equivalent positive-definiteness conditions

For real symmetric S, the following conditions are equivalent.

1. $x^T S x > 0$ for every $x \neq 0$.
2. Every eigenvalue is positive.
3. Every leading principal minor is positive.
4. Symmetric elimination without swaps has positive pivots.
5. $S = A^T A$ for some full-column-rank A.
6. $S = RR^T$ for an invertible lower-triangular R with positive diagonal, the Cholesky factorization.

Symmetric elimination completes squares, giving $S = LDL^T$ and $x^T S x = (L^T x)^T D (L^T x)$. Positive pivots are therefore equivalent to positive energy. Leading minors satisfy $\Delta_k = d_1 \cdots d_k$, so positivity of all leading minors is equivalent to positive pivots. Taking $R = LD^{1/2}$ gives Cholesky, and $A = R^T$ gives the Gram form. Conversely, squared lengths give positive energy. A positive determinant alone is insufficient; $-I_2$ has determinant 1 but is negative definite.

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Checking positive definiteness in dimension two

2×2 :

$$S = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$$

is positive definite exactly when $a > 0, ac - b^2 > 0$.

For example, $S = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ has leading minors 2 and 3, and eigenvalues 1 and 3, so it is positive definite. By contrast, $T = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$ has determinant -3 and is not positive definite.

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Energy proof and counterexamples

$$x^T S x = \sum_i \lambda_i (q_i^T x)^2.$$

If all $\lambda_i > 0$, every nonzero x has some $q_i^T x \neq 0$, making the energy positive. Conversely, an eigenvalue $\lambda_i \leq 0$ supplies a nonzero eigenvector with nonpositive energy, contradicting strict positive definiteness.

For a counterexample,

$$S_1 = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$$

eigenvalues $3, -1$，not positive definite。

If A has independent columns,

$$x^T A^T A x = \|Ax\|^2 > 0$$

for every $x \neq 0$, so $A^T A$ is positive definite. Full row rank cannot replace column independence.

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Positive semidefinite boundary

$$S = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}.$$

At $x = (1, -1)^T$, $x^T S x = 0$; the eigenvalues are $2, 0$. Thus the matrix is positive semidefinite but not positive definite.

For $S = A^T A$, dependent columns give nonzero x with $Ax = 0$ and zero energy. Conversely, zero energy forces $Ax = 0$.

A semidefinite matrix can have an LDL^T factorization, but Cholesky must handle zero pivots. A pseudoinverse does not make a semidefinite matrix invertible.

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Ellipsoids and conditioning

Positive definite S:

$$x^T S x = 1$$

is an origin-centered ellipsoid. In eigen-coordinates,

$$y = Q^T x,$$

$$\sum_i \lambda_i y_i^2 = 1.$$

The semiaxis length in direction i is

$$1/\sqrt{\lambda_i}.$$

The smallest eigenvalue gives the longest semiaxis. Also,

$$\kappa_2(S) = \frac{\lambda_{\max}}{\lambda_{\min}}.$$

Positive definiteness guarantees exact invertibility, but a small minimum eigenvalue relative to the largest still gives poor conditioning.

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Chapter review

- Algebraic multiplicities and eigenspace dimensions determine diagonalizability.
- Jordan chains explain generalized eigenvectors and polynomial factors in powers.
- Asymptotic stability does not preclude transient growth; distinguish one trajectory from an operator norm.
- Real symmetric and complex Hermitian matrices use transpose and conjugate transpose, respectively.

References and attribution

Lin Cing, handwritten linear algebra notes. Gilbert Strang, Introduction to Linear Algebra, 5th ed., Chapter 6.

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